

PowerShell Active Directory & Admin Cheat Sheet

A Dark-Themed Reference Guide for Administrators & Security Professionals

1. ACTIVE DIRECTORY DOMAIN SERVICES (AD DS)

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
<code>Get-ADDomain</code>	Retrieves information about the current Active Directory domain.	Identifies domain controllers, forest details, and domain functional level.
<code>Get-ADUser</code>	Queries user accounts. <code>-Filter *</code> <code>-Properties *</code>	Used to hunt for weak policies, service principal names (Kerberoasting), or stale accounts.
<code>New-ADUser</code>	Creates a new AD user account. <code>-Name "John"</code> <code>-Enabled \$true</code>	Common administrative task; also used by adversaries for persistence.
<code>Get-ADGroupMember</code>	Retrieves members of an AD group. <code>-Identity "Domain Admins"</code>	Crucial for auditing high-privilege groups. Use <code>-Recursive</code> for nested groups.
<code>Add-ADGroupMember</code>	Adds users or groups to an existing AD group.	Requires Domain Admin or delegated administrative permissions.
<code>Get-ADComputer</code>	Queries computers in the domain directory.	Useful for compiling a target list or finding specific server OS types.
<code>Get-ADTrust</code>	Gets all trusted relationships for the domain.	Essential for cross-domain lateral movement & forest mapping.

2. COMPUTER ADMINISTRATION & SYSTEM INFO

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
<code>Get-ComputerInfo</code>	Gets comprehensive system, OS, and hardware details.	The modern replacement for <code>systeminfo</code> . Run as Admin for full output.
<code>Get-Service</code>	Lists local services, status, and startup types.	Identify non-default running services that could be weak or exploit-vulnerable.
<code>Start-Service</code> <code>Stop-Service</code>	Starts or stops a specified system service.	Often used to restart security software or invoke a custom service payload.
<code>Get-Process</code>	Retrieves list of active processes. (gps)	Audit for unauthorized executables, parent-child process anomalies, or EDR/AV processes.

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Stop-Process	Forcefully terminates a process. -Name "notepad" -Force	Kill target security tasks (requires high-privilege context).
Get-HotFix	Returns list of installed security updates.	Cross-reference with CVE databases to identify local kernel exploit vectors.

3. LOCAL ACCOUNTS & SECURITY MANAGEMENT

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
Get-LocalUser	Lists local user accounts on the machine.	Audit for unmanaged local accounts or unauthorized backup admins.
New-LocalUser	Creates a new local user account.	Establish local persistence. Run in elevated prompt.
Get-LocalGroupMember	Lists members of a specific local group. -Group "Administrators"	Identify who has administrative control over the local node.
Add-LocalGroupMember	Adds a local or domain user to a local group.	Promote standard users to local administrators.

4. NETWORK CONFIGURATION & DIAGNOSTICS

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
Get-NetIPAddress	Retrieves IP addresses configuration.	Detailed networking replacement for ipconfig .
Get-NetTCPConnection	Displays active TCP connections and listening ports.	Alternative to netstat . Find listening ports and associated PIDs.
Test-Connection	Sends ICMP echo requests to targets. (ping)	Perform quick network scanning and discovery of live hosts.
Test-NetConnection	Advanced diagnostics (TCP port checks, routing tracer). -ComputerName Target -Port 3389	Incredibly useful built-in utility for ad-hoc port scanning and checking firewall allowances.
Resolve-DnsName	Performs a DNS query. (nslookup replacement)	Gather infrastructure info, find MX, TXT, and SRV records.

5. LOCAL FILE SYSTEM & DATA MANAGEMENT

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
Get-ChildItem	Lists directory items and files. (ls, dir) -Recurse -Hidden	Locate hidden data, configuration scripts, and sensitive file structures.
Get-Content	Reads file contents. (cat) -Tail 50 -Wait	Read scripts or dynamically monitor system/security logs.
Copy-Item	Copies items from one location to another. (cp)	Transfer configuration profiles, payloads, or backup logs.
Remove-Item	Deletes files and folders. (rm) -Recurse -Force	Clean up operational artifacts and temporary tools.
Select-String	Searches files for text matching regex patterns. (grep)	Search credentials: Select-String -Path *.xml -Pattern "password"

6. REMOTE MANAGEMENT & WMI/CIM

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
Invoke-Command	Runs commands on local and remote computers. -ComputerName RemoteHost -ScriptBlock { ... }	The ultimate lateral movement command utilizing WinRM. Supports parallel execution.
Enter-PSSession	Starts an interactive PowerShell session with a remote host.	Allows 1-to-1 administrative CLI access across the network using Kerberos/NTLM authentication.
Get-CimInstance	Gets CIM/WMI instances from local or remote hosts.	Enumerate operating system properties, BIOS, or running processes over standard remote RPC.
Invoke-CimMethod	Invokes a method of a WMI class/CIM instance.	Can be used to execute arbitrary processes remotely via WMI (e.g., executing payload as SYSTEM).

7. UTILITY & PIPELINE ESSENTIALS

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
Get-Help	Displays syntax and usage information for cmdlets.	Use -Examples to understand complex structures quickly.
Get-Command	Lists all cmdlets, functions, and aliases installed.	Identify available administrative modules (e.g., AD, Exchange).

Cmdlet (Verb-Noun)	Description & Common Parameters	Security / Practical Note
Select-Object	Selects specific properties of an object. (select)	Format outputs clean: <code>Get-Process Select-Object ProcessName, Id</code>
Where-Object	Filters objects from the pipeline. (where, ?)	Filter down results: <code>Get-Service Where-Object {\$_.Status -eq "Running"}</code>
Export-Csv	Exports objects to a CSV flat file. <code>-NoTypeInfoation</code>	Easily parse out domain objects, users, or active process tables for report writing.

⚠ Administrator Privilege Notice

Many of the cmdlets listed above (specifically Active Directory querying/management, CIM/WMI methods, local group/user creations, and service manipulations) require execution from an elevated shell context (Run as Administrator) or corresponding Domain Admin privileges.